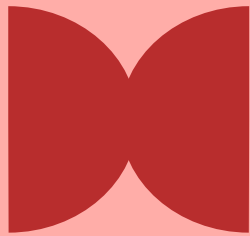


Is a hot dog a
sandwich? A
quantitative
study.



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Motivation: to settle a long standing debate constantly salient in the zeitgeist—is the hot dog a sandwich?



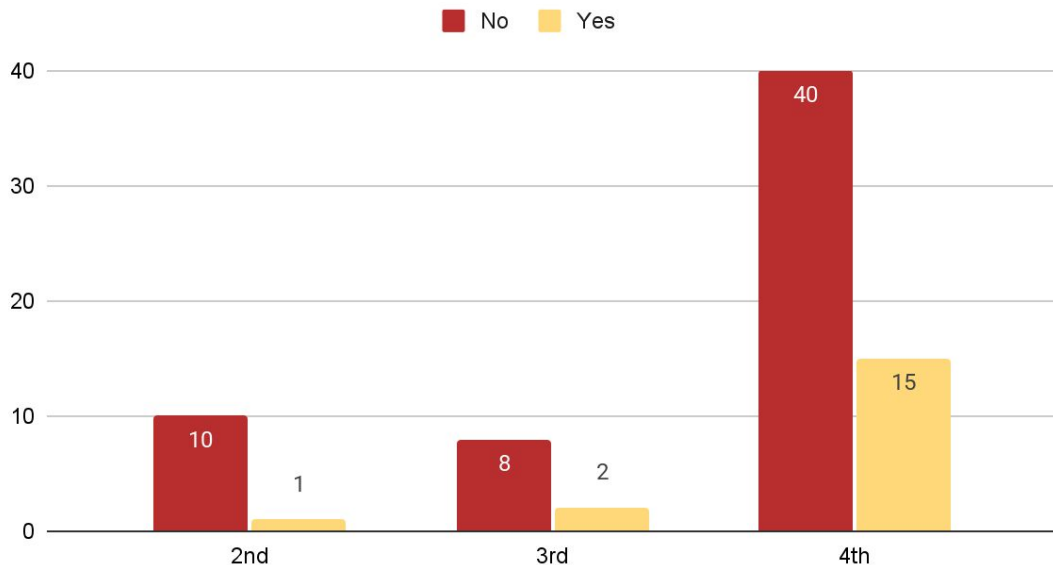
Hypothesis

Less than 50% of full-time undergraduate students at the University of Virginia believe a hot dog is a sandwich.



The Data:

Is a hot dog a sandwich?



79 UVA students were polled, of whom 76 were full-time, undergraduate students.

11 respondents were second years, 10 were third years, and 55 were fourth years.



Analysis Plan

Pre Analysis Steps

- Clean data
 - Exclude students who are not full-time undergraduate students

Analysis Methods

- Pivot tables and charts to count responses
- Z-test to verify results on a larger set of data
- Chi-squared test to test if students' year affects proportion of yes/no responses

Results

Overall, ~23.7% of full-time undergraduate UVA students believe a hot dog is a sandwich.

By year:

- ~9.1% of second years believe a hot dog is a sandwich,
- 20% of third years,
- and ~27.3% of fourth years.

Z score of -4.59 and p-value of 2.23×10^{-6} implies very strong evidence these results apply to the greater population.

Chi squared test p value of 0.46 implies year and belief whether a hot dog is a sandwich are not correlated.

Next Steps

1

Collect more data.

Poll more of the full time UVA undergraduate student population.

2

Re-run the tests.

See how the results of this limited sample compare to the greater population.

3

Ask other people.

Examine how other populations feel about this issue.

4

Is cereal a soup?

Ask another question!